

LI-HSIANG CHAO

趙立翔

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Education

University of Colorado Boulder — M.S. Data Science, 2024-2026, GPA 3.8/4.0

Statistical modeling and inference | Database design and SQL | Data mining | Machine learning |
Computer vision | Neural network algorithm | Robotics and autonomous systems

Tamkang University — M.S. Information Management, 2023-2026, Scholarship Recipient, GPA 4.0/4.0

Thesis: A symbolic regression approach for clinical trial eligibility criteria optimization using EHR

This study presents a machine learning-based, multi-objective optimization framework for clinical trial cohort selection that, validated on both benchmark and real-world electronic health record data, outperforms the existing benchmark approach in treatment-effect estimation while offering greater interpretability in balancing efficacy with population inclusivity.

National Yang Ming University — B.S. Life Science, 2012-2016

Work Experience

Microbio group and Oneness Biotech Co., Ltd. (TW 4743), Oct. 2017 – Present

Research Assistant & ISO 9001 Internal Quality Auditor

- Developed machine learning-based analytical pipelines for RNA-seq clinical data, enabling more efficient identification of drugs responder.
- Conducted data analysis and statistical evaluation to support FB704A and FB825 clinical projects.
- Contributed to licensing negotiation preparation for FB825 with Leo Pharma by developing a flow cytometry assay for identifying membrane-bound IgE⁺ B cell.
- Managed patent-related documentation workflow, ensuring compliance and timely filing.

Research Works and Awards

- Li-Hsiang Chao and C.-B. Cheng, 2026, Clinical Trial Eligibility Criteria Optimization based on Electronic Health Records: A Machine-Learning Framework Powered by Symbolic Regression, 2026 International Conference on Information Management (ICIM2026), May 16, 2026, Taipei, Taiwan
 - **Distinguished Award (2nd place)**, Y. S. Educational Foundation AI Digital Technology Thesis Award
 - Selected as **the best paper** of ICIM2026
- Algorithmic Trading Systems
 - Developed an algorithmic trading bot in MQL4, implementing multiple quantitative trading strategies.
 - Designed a modular architecture integrating financial products databases and financial information APIs to automate trading signal generation, risk control, and order execution.
- FathomNet ML image classification competition — Top 25%, Kaggle
 - Developed transformer models for unknown marine image classification and performance evaluation.

Proficiencies & Certifications

TOEFL iBT 94 | Python, R, C, SQL | Git, Docker | AI-assisted coding, Sequential and parallel computing

GenAI: LLMs, RAG, Graph RAG, local LLM-RAG workflows

Data Engineering & Cloud: Azure, relational database design, SQL-based data processing

Robotics & Modeling: Robot control, Odometry, Mapping, Path planning, 3D modeling

Statistics: Hypothesis testing, Generalized linear models, Generalized additive models, Model diagnostics

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